

LEARN BY DOING

Your path. Your practice.

A practical guide to choosing exercises, checking your work and building evidence for a data or AI career.

What Practice Lab does

Your career, selected tools and industry shape the exercises you see. Work with synthetic data, receive feedback and return to the relevant lesson whenever you need help.

1

Set your direction

Choose a career and business sector during setup. Add the tools you want to learn. These choices shape your practice catalogue.

2

Open Practice Lab

Start with **Your next exercise**. Its explanation tells you why it was recommended. You can also filter by skill, browse the catalogue or download this guide from the Practice Lab header.

3

Read the task and inspect the data

Check the business rule, field names and units. Use the difficulty and estimated time to choose a manageable session.

4

Try, check and improve

Run your query or code, check a formula, calculate a metric, or save a case submission. Review feedback and reveal hints one at a time.

5

Keep the evidence

Revisit saved attempts, restore an earlier answer and improve it. Successful checked exercises add evidence to their related lessons.

AVAILABLE NOW

Six ways to practise

Only relevant exercises appear for your path. Tool interests can add practice in additional skills.

SQL | Run queries

Query a synthetic industry dataset. Practise filters, aggregation, joins, CTEs, ranking and debugging.

Check: real results plus three alternate datasets. Up to 12 challenges, depending on your target proficiency.

Python | Run code

Write a **solve(rows)** function for totals, missing values or an average that needs debugging.

Check: four datasets, including changed values, empty input and missing-data edge cases.

Excel | Formula exercise

Practise a conditional total with **SUMIF**, using the status and value columns shown in the task.

Scope: a focused formula checker, not a complete Excel workbook.

DAX | Measure exercise

Define a filtered measure using **CALCULATE** and **SUM** for the provided events.

Scope: a focused expression exercise, not the full Power BI engine.

Metrics | Calculate

Define the numerator and denominator, then calculate a result such as a completion rate.

Check: a numerical answer with feedback about the calculation.

Applied cases | Build evidence

Complete a relevant case or industry mission. Check three reconciliation calculations, then explain your method. Download CSV data for your tools.

Review: save artifact links, reflections and a self-review checklist.

Code and calculation checks provide practice feedback. A self-reviewed case is not an independently verified qualification.

MAKE EACH ATTEMPT COUNT

From task to evidence

Keep the business question in view while you work. The desktop workspace places reference material beside your answer; smaller screens stack the panels.

Reference side

Read the task and lesson explanation. Inspect the dataset and download a CSV where offered. Reveal one hint, make another attempt, then ask for the next hint if needed.

Answer and feedback side

Enter your query, code, formula or response. Run the check and inspect the result. Follow the related-lesson link when a requirement is unclear.

Understand the outcome

Passed: your answer met the exercise checks.

Needs work: inspect the failed requirement, correct the answer and retry.

Submission saved: your case is stored for self-review; it has not been reviewed by an expert.

Your industry changes the context

Retail uses stores and orders; healthcare uses clinics and claims; telecom uses regions and sessions. Other choices include banking, finance, marketing, technology, government and general. Read the stated units and limitations before interpreting any result.

Keep useful work

The latest **50 SQL attempts** and **30 other attempts or case revisions** are retained. Pass records are stored separately. SQL passes are separated by industry; other exercise passes also reflect your career. Use the existing progress-save and export options for your account or guest session.

If an exercise gets stuck

Python may take up to 45 seconds to load initially. Code execution has an eight-second limit. Simplify long-running code and retry. For SQL errors, check field names, quotes and clause order. Excel and DAX accept only the formula forms described in their tasks.

PRODUCT ROADMAP

Available now. Coming next.

The guide distinguishes working features from planned improvements. Upcoming items have no committed release date.

NOW | A clearer starting point and workspace

Next-exercise recommendation: uses your path, unfinished topics, review queue and recent attempts, with a visible reason.

Focused workspace: keeps the catalogue collapsible, places reference material beside the editor on larger screens, and adds difficulty, estimated time and progressive hints.

01 Expand the industry missions

Available now: a three-checkpoint reconciliation mission in each industry, with duplicates, late corrections and missing values. Planned next: larger datasets and more varied scenarios.

02 Deeper practice tools

Planned: Python notebooks with pandas, broader spreadsheet exercises and multi-step SQL projects. The current Python exercises and formula checkers are the starting point.

03 Stronger assessment and project review

Planned: richer solution explanations, more targeted follow-up exercises and reviewer feedback on submitted projects. Independent review will remain distinct from self-review.

Help shape the next release

When giving feedback, include your career, industry, exercise title, what you expected and what happened. Mention whether you were using a phone or a computer. Avoid sharing private or sensitive data.